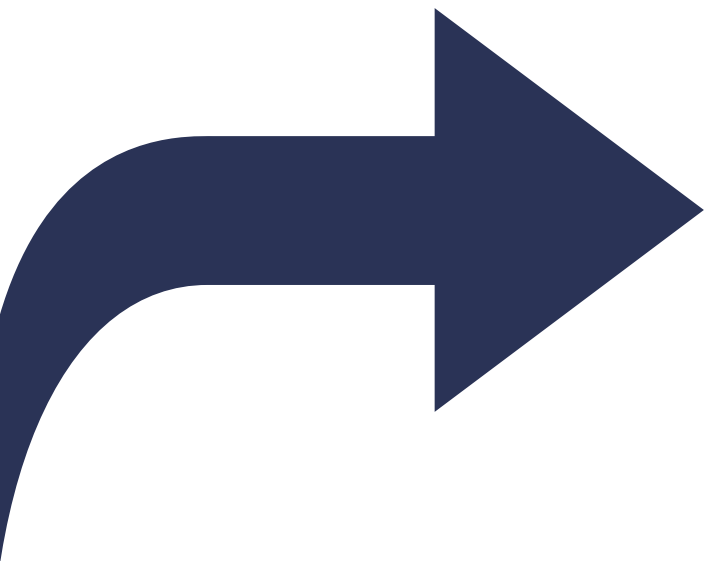


Introducción al modelo OMOP-CDM

Alberto Labarga – Febrero 2026





I am Alberto



alabarga



alabarga



/in/albertolabarga



alberto.labarga@gmail.com

Febrero
2026 

OMOP-CDM

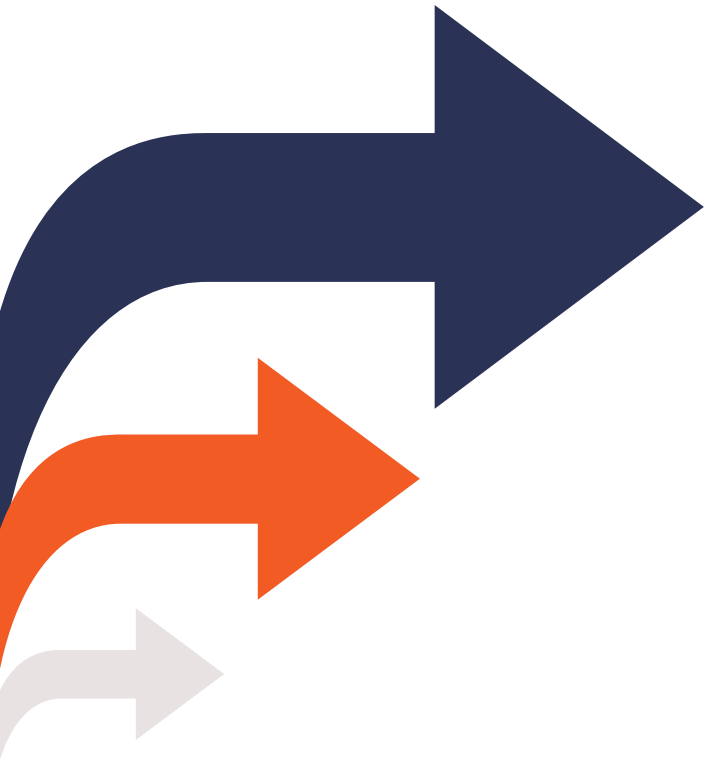
El curso

El curso tiene como objetivo proporcionar una comprensión profunda del modelo de datos **OMOP-CDM** de la OHDSI, incluida su **estructura**, tablas clave y **vocabularios**.

Abordaremos también como **transformar** datos al modelo y explorar y analizar la **calidad** de los datos transformados.

L M X J V S D						
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	1
2	3					

Objetivos del curso



01

Introducción al modelo OMOP-CDM, a sus tablas y sus vocabularios

02

Transformación de datos al modelo OMOP-CDM utilizando la suite de herramientas de la OHDSI

03

Análisis de calidad y explotación de una base de datos OMOP-CDM

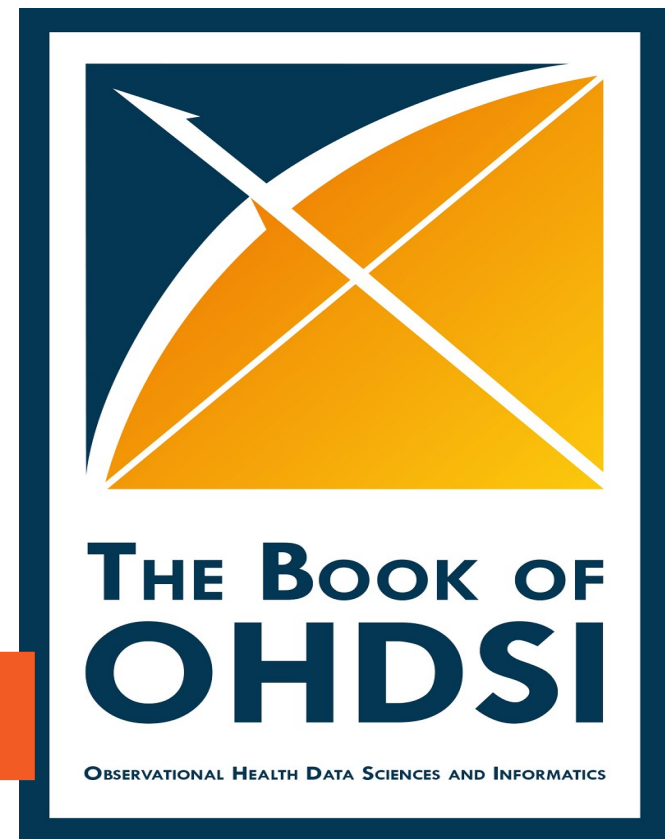
Recursos de aprendizaje



Sitio web: <https://github.com/alabarga/curso-omop-cdm>



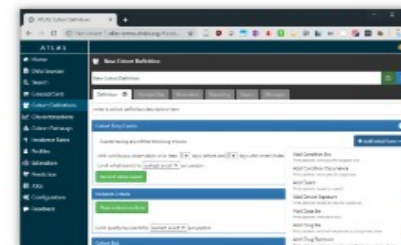
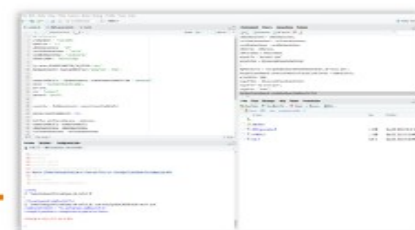
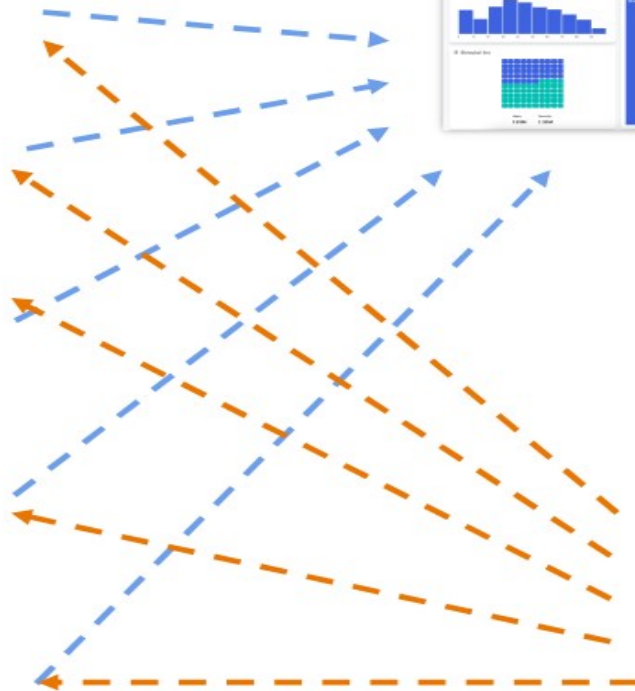
Libro <https://ohdsi.github.io/TheBookOfOhdsi/>





OMOP-CDM^{*}

VOCABULARIOS

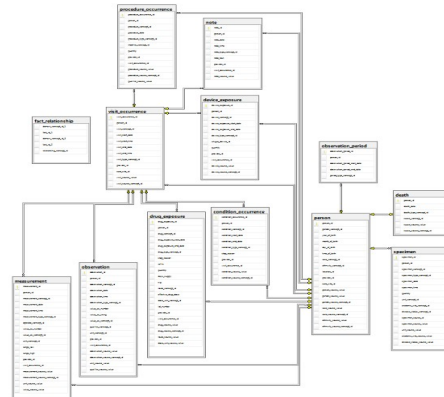


OMOP-CDM

Common Data Model

OMOP-CDM pretende servir como modelo de datos común para poder comprender los datos clínicos sin importar su origen. Este es un estándar de **licencia abierta** cuyo objetivo es estandarizar los datos y estructurarlos para permitir análisis y observaciones eficientes cuyos resultados sean fiables.

Propone **interoperabilidad sintáctica** (modelo de datos) e **interoperabilidad semántica** (vocabularios)



ATHENA

SEARCH BY KEYWORD: cancer

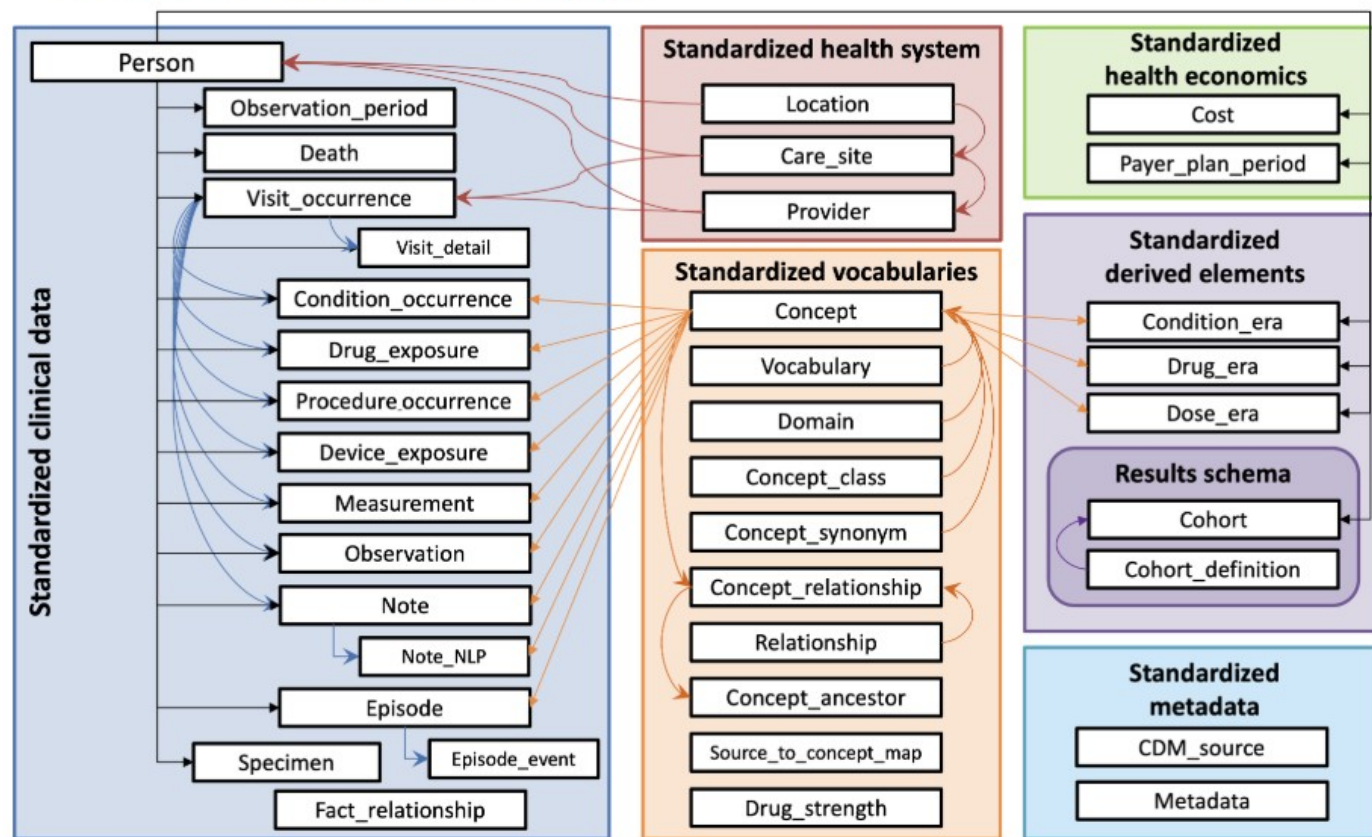
DOWNLOAD RESULTS

Show by 15 items Total 8,507 items

ID	CODE	NAME	CLASS	CONCEPT	VALIDITY	DOMAIN	VOCAB
45877275	LA18524-9	Cancer	Answer	Standard	Valid	Meas Value	LOINC
1194405	38000000	Cancer confirmed	Control dependent	Standard	Valid	Condition	SNOMED
4299008	38000001	Cancer register	Source Context	Standard	Valid	Observation	SNOMED
1384533	CancerCardInfo_H3UCancer	Cancer Conditions: No Cancer	Answer	Standard	Valid	Observation	PR
4873814	343073306	Cancer education	Procedure	Standard	Valid	Procedure	SNOMED
44803809	722721000000100	Colorectal cancer	Qualifier Value	Standard	Valid	Observation	SNOMED
42157663	54520-7	Cancer	Survey	Standard	Valid	Observation	LOINC

The current CDM version is [CDM v5.4](#), depicted below. This CDM version was developed over the course of a year by considering requests that were sent via our [issues page](#). The list of proposed changes was then shared with the community in multiple ways: through discussions at the weekly OHDSI Community calls, discussions with the OHDSI Steering Committee, and discussions with all potentially affected workgroups. The [final changes](#) were then delivered to the Community through a new R package designed to dynamically generate the DDLs and documentation for all supported SQL dialects. Looking for an entity-relationship diagram? Click [here](#)!

- [Link to DDLs for CDM v5.4](#)
- [Link to ReadMe for instructions on how to use the R package](#)



[CommonDataModel](#) / [inst](#) / [ddl](#) / [5.4](#) / [🔗](#)

[clairblacketer](#) Removing codespell and older ddl folder for CRAN submission

Name
..
bigquery
duckdb
hive
impala
netezza
oracle
pdw
postgresql
redshift
snowflake
spark
sql_server
sqlite
sqlite_extended
synapse

Show all



SHOW HISTORY

DOWNLOAD VOCABULARIES

☐	ID (CDM V4.5)	CODE (CDM V5)	NAME	REQUIRED	LATEST UPDATE
☒	1	SNOMED	Systematic Nomenclature of Medicine - Clinical Terms (IHTSDO)		01-Mar-2025
☒	2	ICD9CM	International Classification of Diseases, Ninth Revision, Clinical Modification, Volume 1 and 2 (NCHS)		01-Oct-2014
☒	3	ICD9Proc	International Classification of Diseases, Ninth Revision, Clinical Modification, Volume 3 (NCHS)		01-Oct-2014
☒	4	CPT4	Current Procedural Terminology version 4 (AMA)	EULA required	05-May-2025
☒	5	HCPCS	Healthcare Common Procedure Coding System (CMS)		01-Jul-2025
☒	6	LOINC	Logical Observation Identifiers Names and Codes (Regenstrief Institute)		26-Feb-2025
☐	7	NDFRT	National Drug File - Reference Terminology (VA)		06-Aug-2018
☒	8	RxNorm	RxNorm (NLM)		02-Jun-2025
☒	9	NDC	National Drug Code (FDA and manufacturers)		17-Aug-2025

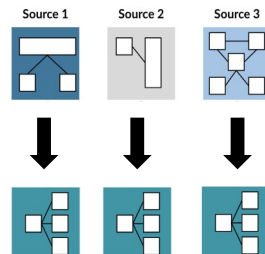
TRANSFORMACIÓN AL MODELO OMOP-CDM



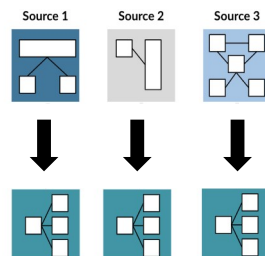
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TRANSFORMACIÓN AL MODELO OMOP-CDM

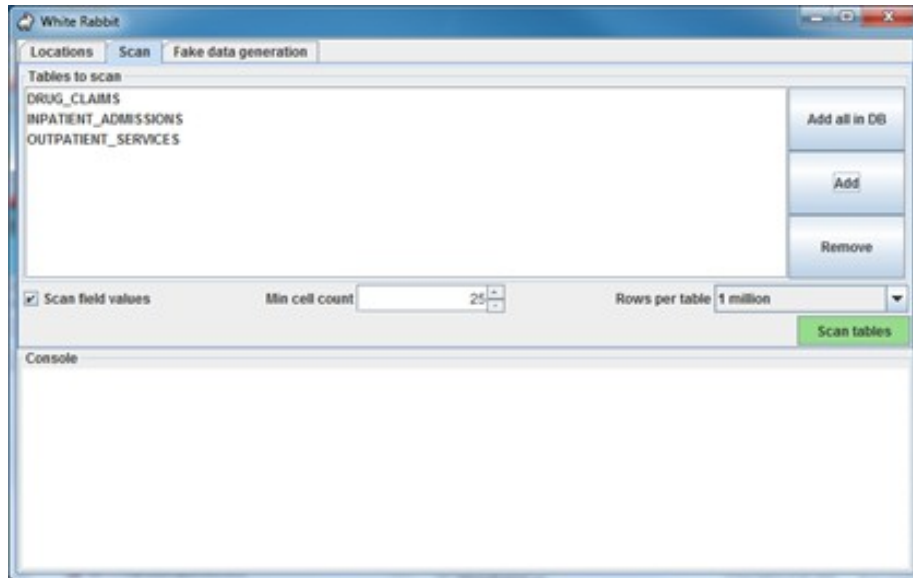


TRANSFORMACIÓN AL MODELO OMOP-CDM



WhiteRabbit

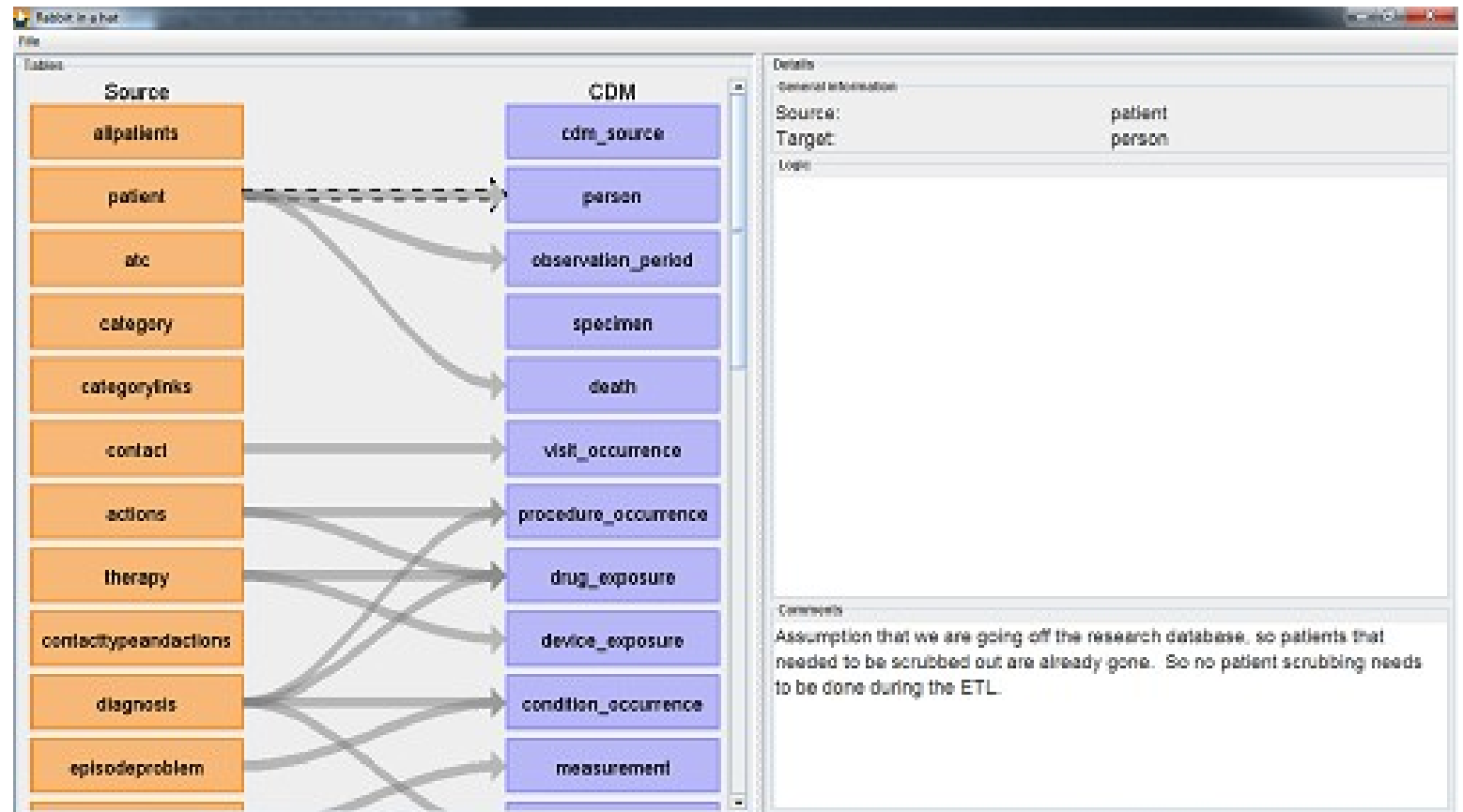
DATA PROFILING



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1	Table	Field	Type	Max length	N rows	N rows checked	Fraction empty	
2	atc	atc	text	7	6231	6231	0	
3	atc	txt	text	58	6231	6231	0	
4	atc	dddhp	double precision	8	6231	6231	0.951693147167389	
5								
6	contact	expatidx	numeric	7	100000	100000	0	
7	contact	prakid	numeric	9	100000	100000	0	
8	contact	patid	numeric	9	100000	100000	0	
9	contact	contactid	numeric	5	100000	100000	0	
10	contact	contactd	date	10	100000	100000	0.00182	
11	contact	dateperiod	text	1	100000	100000	0	
12	contact	rawcodsys	text	6	100000	100000	0.61758	
13	contact	rawcod	text	9	100000	100000	0.30036	
14	contact	rawcodtxt	text	62	100000	100000	0.38449	
15	contact	codsys	text	6	100000	100000	0.39153	
16	contact	cod	text	5	100000	100000	0.39153	
17	contact	codtxt	text	151	100000	100000	0.39153	
18	contact	employeeid	numeric	9	100000	100000	0.50622	
19								

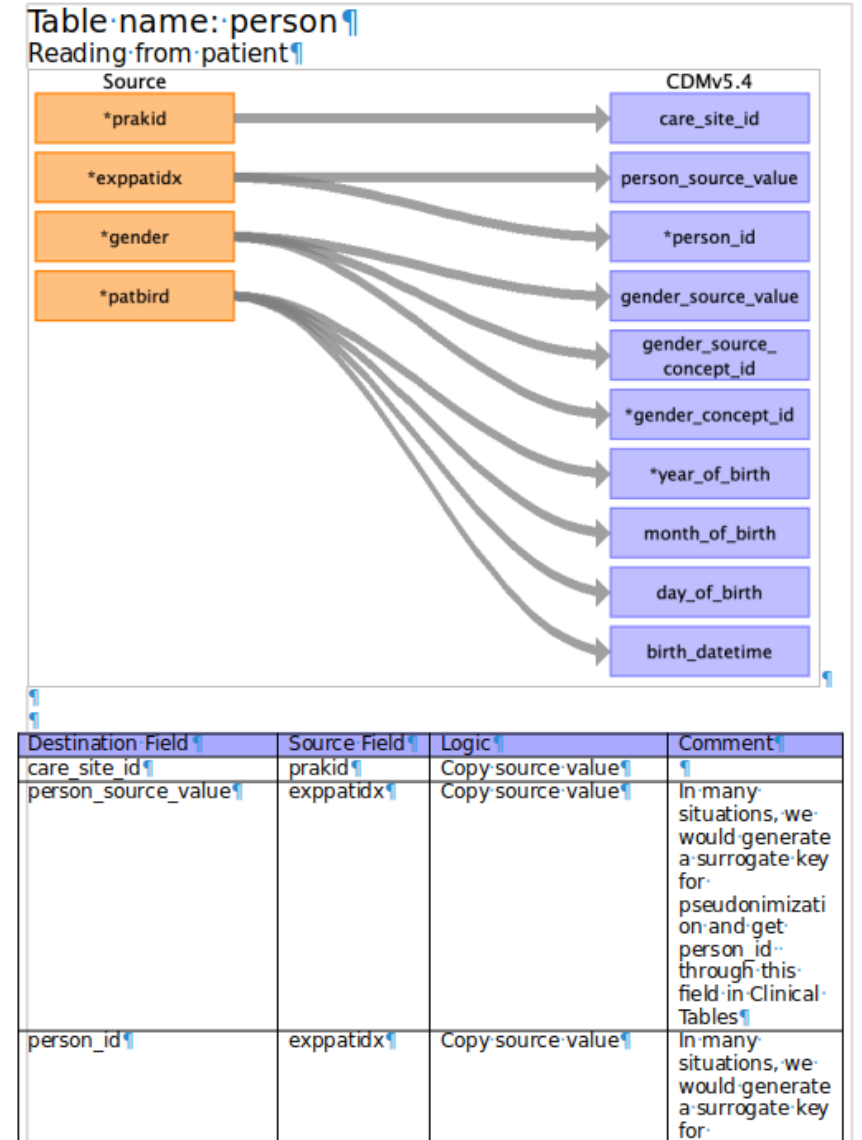
Rabbit -in-a-Hat

DATA MAPPING

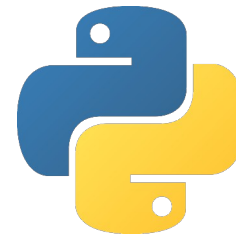
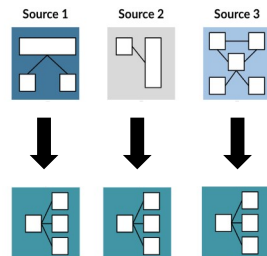


Rabbit -in-a-Hat

DATA MAPPING



TRANSFORMACIÓN AL MODELO OMOP-CDM



dbt

TRANSFORMACIÓN AL MODELO OMOP-CDM

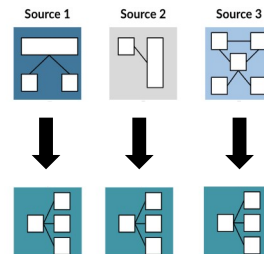
Extract

Transform

Map

Process

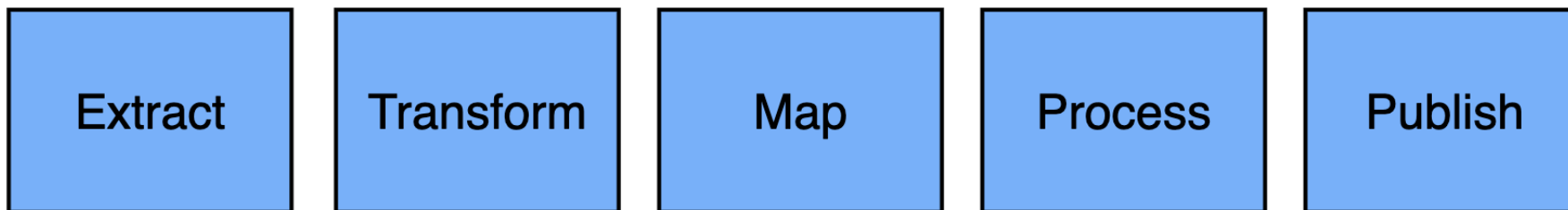
Publish



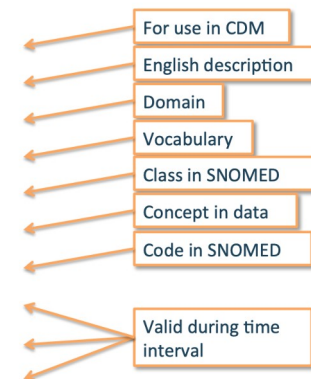
```
select
  {{ create_id_from_str("Id":text)}} AS person_id,
  {{ gender_concept_id ("GENDER") }} AS gender_concept_id,
  date_part('year', "BIRTHDATE"::DATE)::INT AS year_of_birth,
  date_part('month', "BIRTHDATE"::DATE)::INT AS month_of_birth,
  date_part('day', "BIRTHDATE"::DATE)::INT AS day_of_birth,
  "BIRTHDATE"::TIMESTAMP AS birth_datetime,
  {{ race_concept_id("RACE") }} AS race_concept_id,
  {{ ethnicity_concept_id("ETHNICITY") }} AS ethnicity_concept_id,
  NULL::INT AS location_id,
  NULL::INT AS provider_id,
  NULL::INT AS care_site_id,
  "Id"::VARCHAR(50) AS person_source_value,
  "GENDER"::VARCHAR(50) AS gender_source_value,
  @ AS gender_source_concept_id,
  "RACE"::VARCHAR(50) AS race_source_value,
  @ AS race_source_concept_id,
  "ETHNICITY"::VARCHAR(50) AS ethnicity_source_value,
  @ AS ethnicity_source_concept_id
from patients
where "BIRTHDATE" is not null -- Don't load patients who do not have birthdate and sex (change variable
names (if necessary)
and "GENDER" is not null
return go(f, seed, [])
}
```

TRANSFORMACIÓN

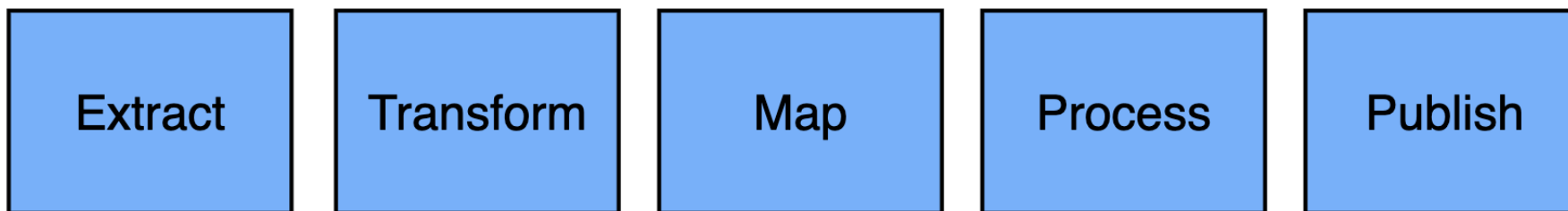
AL MODELO OMOP-CDM



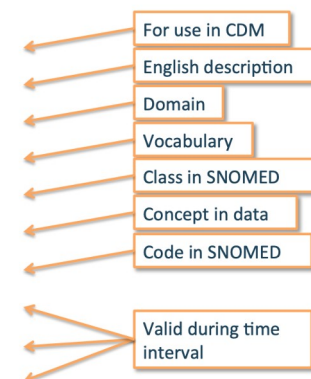
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CONCEPT_NAME	Atrial fibrillation
DOMAIN_ID	Condition
VOCABULARY_ID	SNOMED
CONCEPT_CLASS_ID	Clinical Finding
STANDARD_CONCEPT	S
CONCEPT_CODE	49436004
VALID_START_DATE	01-Jan-1970
VALID_END_DATE	31-Dec-2099
INVALID_REASON	



TRANSFORMACIÓN AL MODELO OMOP-CDM



CONCEPT_ID	313217
CONCEPT_NAME	Atrial fibrillation
DOMAIN_ID	Condition
VOCABULARY_ID	SNOMED
CONCEPT_CLASS_ID	Clinical Finding
STANDARD_CONCEPT	S
CONCEPT_CODE	49436004
VALID_START_DATE	01-Jan-1970
VALID_END_DATE	31-Dec-2099
INVALID_REASON	



Usagi

CONCEPT MAPPING

Status	Source code	Source term	Frequency	CodeText	Match score	Concept ID	Concept name	Domain	Concept class	Vocabulary	Concept code	Standard con...	Parents	Children	Comment
Approved	K87.00	Hypertension...	694195	Hypertensie ...	0.81	316866	Hypertensive...	Condition	Clinical Findi...	SNOMED	38341003	S	1	27	
Approved	L99.00	Other diseas...	680422	Andere ziekte...	0.47	0	Unmapped						0	0	Too generic
Approved	D01.00	Abdominal p...	678588	Gegeneralis...	0.61	197988	Generalized ...	Condition	Clinical Findi...	SNOMED	102614006	S	1	0	
Unchecked	S99.00	Skin disease...	675817	Andere ziekte...	0.75	4317258	Disorder of s...	Condition	Clinical Findi...	SNOMED	95320005	S	2	193	
Unchecked	T86.00	Hvnothvroidi	667283	HvnothvrenA	1.00	4113642	Hvnothvroidi	Condition	Clinical Findi...	SNOMED	286910004	S	1	0	

Source code	Source term	Frequency	CodeText
S99.00	Skin disease, other	675817	Andere ziekte(n) huid/subcutis

Target concepts	Concept ID	Concept name	Domain	Concept class	Vocabulary	Concept code	Standard concept	Parents	Children
4317258	Disorder of skin	Condition	Clinical Finding	SNOMED	95320005	S	2		193

Search

Query

☒ Use source term as query

☐ Query:

Filters

☐ Filter by user selected concepts

☒ Filter standard concepts

☒ Include source terms

☐ Filter by concept class:

☐ Filter by vocabulary:

☒ Filter by domain:

2-dig nonbill code

ABMS

Condition

Results	Score	Term	Concept ID	Concept name	Domain	Concept class	Vocabulary	Concept code	Standard concept	Parents	Children
0.75	Skin disease	4317258	Disorder of skin	Condition	Clinical Finding	SNOMED	95320005	S	2		193
0.65	Skin Disease, Fungal	137213	Dermal mycosis	Condition	Clinical Finding	SNOMED	14560005	S	3		12
0.57	AIDS with skin dise...	4224566	Skin disorder associated with AIDS	Condition	Clinical Finding	SNOMED	421394009	S	2		2
0.56	Chronic skin disease	4134132	Chronic disease of skin	Condition	Clinical Finding	SNOMED	128236002	S	2		26
0.55	Disease, Otologic	378161	Disorder of ear	Condition	Clinical Finding	SNOMED	25906001	S	4		43
0.55	Disease, Hers	4163346	Glycogen storage disease, type VI	Condition	Clinical Finding	SNOMED	29291001	S	2		0
0.55	Other peripheral va...	321052	Peripheral vascular disease	Condition	Clinical Finding	SNOMED	400047006	S	1		44
0.55	Other peripheral va...	4119612	Lower limb ischemia	Condition	Clinical Finding	SNOMED	233961000	S	2		3
0.55	Disease, Ormond	4176725	Retroperitoneal fibrosis	Condition	Clinical Finding	SNOMED	49120005	S	1		3
0.54	Pathological fractur...	73571	Pathological fracture	Condition	Clinical Finding	SNOMED	268029009	S	1		21
0.52	Disease, Tooth	4122115	Tooth disorder	Condition	Clinical Finding	SNOMED	234947003	S	3		58
0.52	Disease, Lip	135858	Disorder of lip	Condition	Clinical Finding	SNOMED	90678009	S	3		35
0.51	Disease, Ollier	4113600	Multiple congenital exostosis	Condition	Clinical Finding	SNOMED	254044004	S	6		0

Replace concept

Add concept

Comment:

Approved / total: 5 / 1037 8.8% of total frequency

Approve

TRANSFORMACIÓN

AL MODELO OMOP-CDM

Extract

Transform

Map

Process

Publish

El paciente acude con su madre **Pepita PER**, con la que vive en **Calle Córcega 23 LOC**, refiriendo dolor abdominal.

El paciente acude con su madre **Lucía PER**, con la que vive en **Calle Londres LOC**, refiriendo dolor abdominal.

TRANSFORMACIÓN AL MODELO OMOP-CDM

Extract

Transform

Map

Process

Publish

Acude con dolor abdominal derecho de 2 días de duración. Pauta de vacunación completa.

Finding *dolor abdominal.*

Spatial Concept *derecho.*

Temporal Concept *2 días, duración.*

Therapeutic or Preventive Procedure *vacunación.*

Qualitative Concept *completa.*

PROCESADO

DEL LENGUAJE NATURAL

Estadísticas generales de los resultados obtenidos con el procesamiento de notas.

- Number of patients with notes: 934.000
- Number of notes: 46.430.000
- Average length: 700
- Average number of entities detected per note: 56

Domain	Events in structure data	Events retrieved by NLP
Condition	2.562.120	537.854.624
Procedure	800.717	205.344.592
Observation	335.659	1.621.526.706
Drug	5.002.550	132.818.091
Measurement	185.752.461	158.167.899
Total	194.4M	2.650M

TRANSFORMACIÓN AL MODELO OMOP-CDM

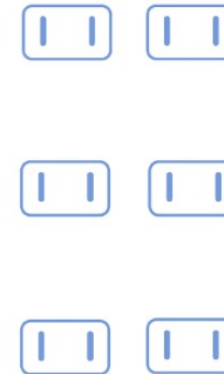
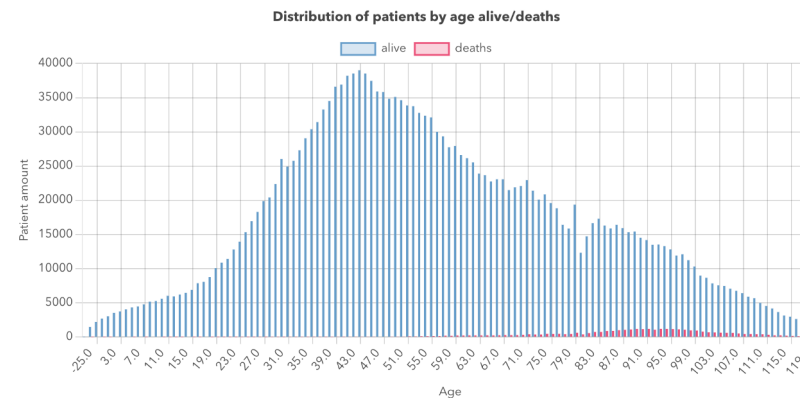
Extract

Transform

Map

Process

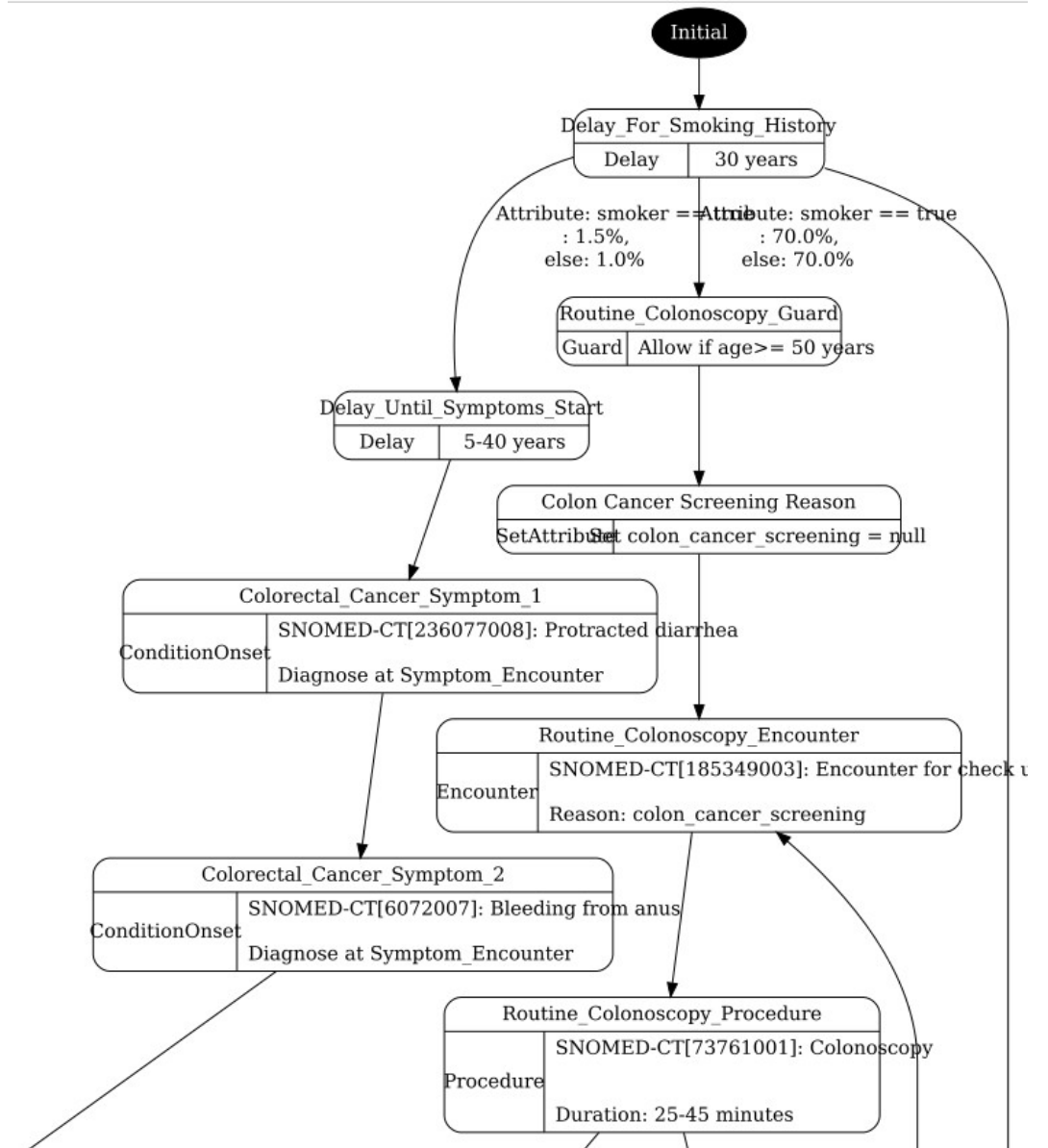
Publish



Ejercicio

Transformación a OMOP-CDM

DATOS SINTÉTICOS

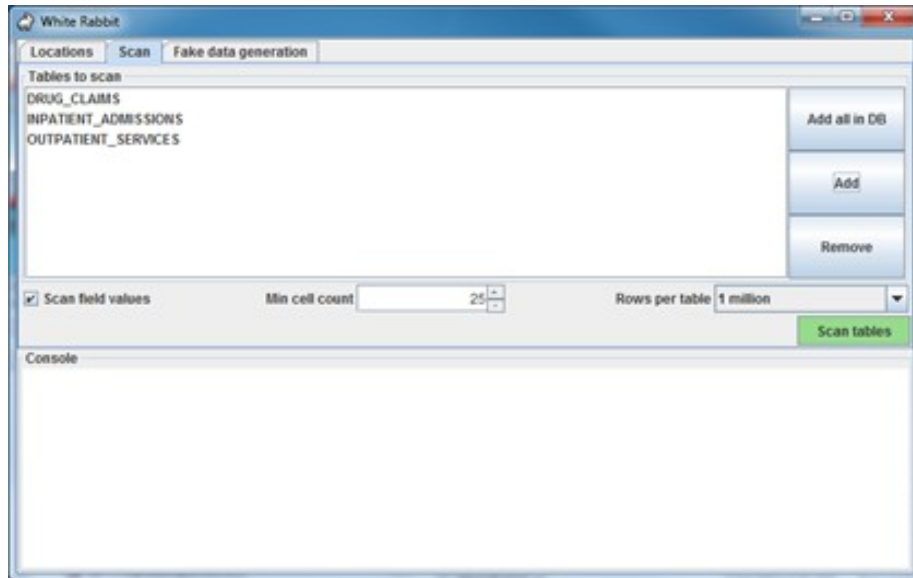


DATOS SINTÉTICOS

File	Description
allergies.csv	Patient allergy data.
careplans.csv	Patient care plan data, including goals.
claims.csv	Patient claim data.
claims_transactions.csv	Transactions per line item per claim
conditions.csv	Patient conditions or diagnoses.
devices.csv	Patient-affixed permanent and semi-permanent devices.
encounters.csv	Patient encounter data.
imaging_studies.csv	Patient imaging metadata.
immunizations.csv	Patient immunization data.
medications.csv	Patient medication data.
observations.csv	Patient observations including vital signs and lab reports.
organizations.csv	Provider organizations including hospitals.
patients.csv	Patient demographic data.
payer_transitions.csv	Payer Transition data (i.e. changes in health insurance).
payers.csv	Payer organization data.
patients_and_conditions.csv	Patient conditions data including diagnoses.

WhiteRabbit

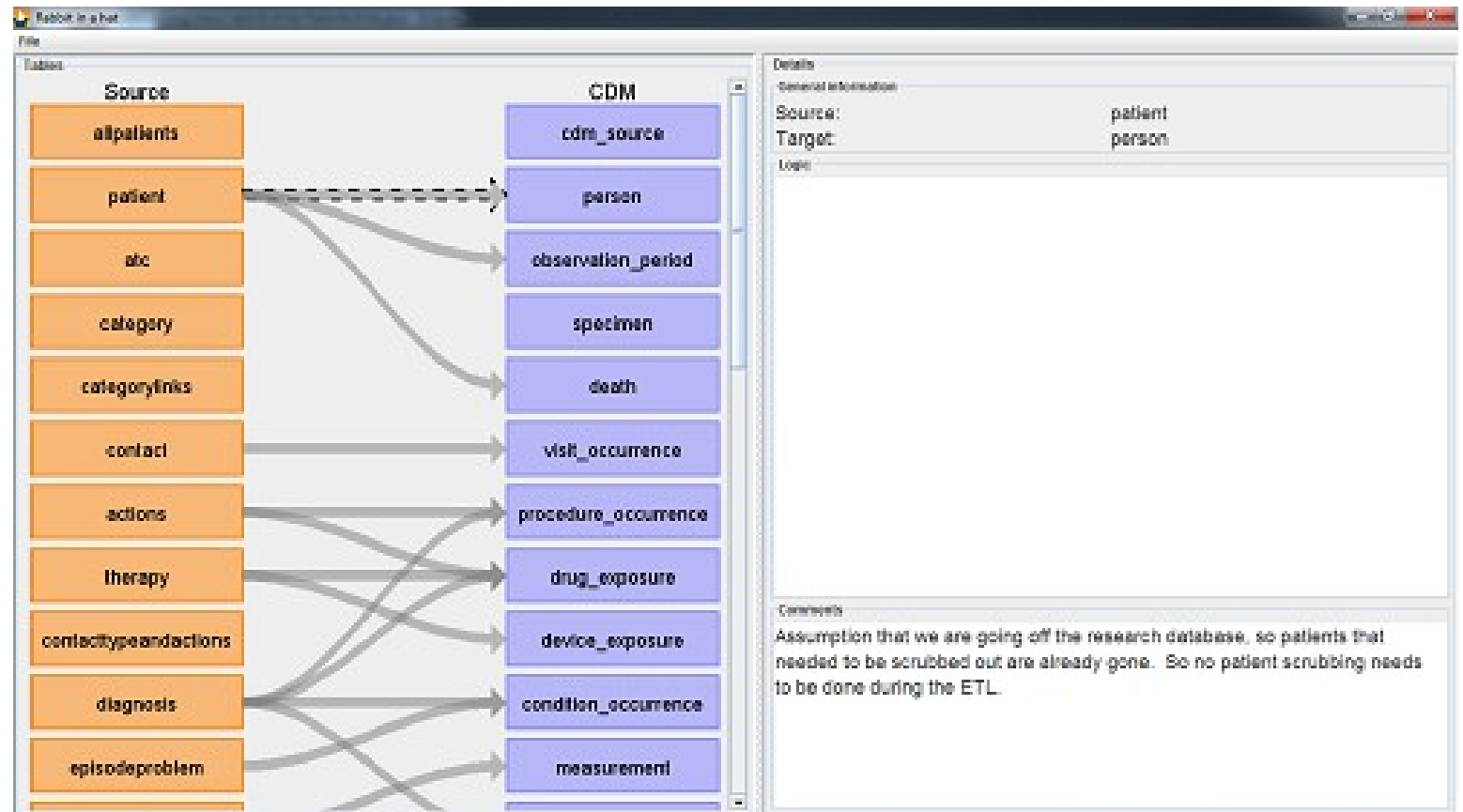
DATA PROFILING



File Edit View Insert Format Styles Sheet Data Tools Window Help							
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	A	B	C	D	E	F	G
1	Table	Field	Type	Max length	N rows	N rows checked	Fraction empty
2	atc	atc	text	7	6231	6231	0
3	atc	txt	text	58	6231	6231	0
4	atc	dddhp	double precision	8	6231	6231	0.951693147167389
5							
6	contact	expatidx	numeric	7	100000	100000	0
7	contact	prakid	numeric	9	100000	100000	0
8	contact	patid	numeric	9	100000	100000	0
9	contact	contactid	numeric	5	100000	100000	0
10	contact	contactd	date	10	100000	100000	0.00182
11	contact	dateperiod	text	1	100000	100000	0
12	contact	rawcodsys	text	6	100000	100000	0.61758
13	contact	rawcod	text	9	100000	100000	0.30036
14	contact	rawcodtxt	text	62	100000	100000	0.38449
15	contact	codsys	text	6	100000	100000	0.39153
16	contact	cod	text	5	100000	100000	0.39153
17	contact	codtxt	text	151	100000	100000	0.39153
18	contact	employeeid	numeric	9	100000	100000	0.50622
19							

Rabbit -in-a-Hat

DATA MAPPING



Usagi

CONCEPT MAPPING

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Unchecked	T86.00	Hvnothvroldi...	667283	HvnothvrenA...	1.00	4113642	Hvnothvroldi...	Condition	Clinical Findi...	SNOMED	286910004	S	1	0	

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☒ Use source term as query

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Filters

☐ Filter by user selected concepts

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☒ Filter by domain:

2-dig nonbill code

ABMS

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0.54	Pathological fractur...	73571	Pathological fracture	Condition	Clinical Finding	SNOMED	268029009	S	1		21
0.52	Disease, Tooth	4122115	Tooth disorder	Condition	Clinical Finding	SNOMED	234947003	S	3		58
0.52	Disease, Lip	135858	Disorder of lip	Condition	Clinical Finding	SNOMED	90678009	S	3		35
0.51	Disease, Ollier	4113600	Multiple congenital exostosis	Condition	Clinical Finding	SNOMED	254044004	S	6		0

Replace concept

Add concept

Comment:

Approved / total: 5 / 1037 8.8% of total frequency

Approve

<https://github.com/OHDSI/Usagi>

Usagi

Mapeo semántico a OMOP-CDM

USAGI

MAPEO SEMÁNTICO

Los códigos fuente que necesitan ser mapeados se cargan en Usagi.

Si los códigos no están en inglés, se requieren columnas adicionales con las traducciones.

Se utiliza un enfoque de similitud de términos para conectar los códigos fuente con los conceptos del Vocabulario.

Hay un paso previo de indexado de los vocabularios descargados de Athena

USAGI

MAPEO SEMÁNTICO

La secuencia típica para utilizar Usagi es:

- Cargar los códigos fuente que deseamos mapear a conceptos del Vocabulario OMOP.
- Utilizaremos la interfaz de Usagi para revisar los mapeos sugeridos o crear nuevos mapeos. Generalmente, esta revisión debe realizarla una persona con experiencia en el sistema de codificación y en terminología médica.
- Exportar el mapeo final generado por Usagi a la tabla SOURCE_TO_CONCEPT_MAP del Vocabulario OMOP.

Import codes from ICPC2SNOMED.csv

Code	English term	Count	UMLS lookup	Dutch term
A99.00	General disease	5774012		Andere gegeneraliseerde/niet gespecif...
K86.00	Hypertension uncomplicated	3987206		Essentiële hypertensie zonder orgaan...
R44.00	Preventive Immunisations/Medications	3702922		Immunisatie/preventieve medicatie
T90.02	Diabetes mellitus type 2	2275799		Diabetes mellitus type 2
R05.00	Cough	1268829	4158493	Hoesten
R74.00	Upper respiratory infection acute	1061504		Acute infectie bovenste luchtwegen
A29.00	General symptom/complaint other	1035167		Andere algemene symptomen/klachten
L03.00	Low back symptom/complaint	998249		Lage rugpijn zonder uitstraling [ex. L86]
U71.00	Cystitis/urinary infection other	970719		Cystitis/urinewegsinfecties
A60.00	Results Tests/Procedures	903897		Uitslag onderzoek/verrichting
R97.00	Allergic rhinitis	892467	257007	Hooikoorts/allergische rhinitis
A97.00	No disease	868585		Geen ziekte
R96.00	Asthma	859684		Astma
P76.00	Depressive disorder	856960	440383	Depressie
P06.00	Sleep disturbance	777549		Slapeloosheid/andere slaapstoornis
R95.00	Chronic obstructive pulmonary disorder	764354	4188974	Emfyseem/COPD
S88.00	Dermatitis contact/allergic	736689		Contact eczeem/ander eczeem
R78.00	Acute bronchitis/bronchiolitis	728202	4110022	Acute bronchitis/bronchiolitis
D12.00	Constipation	723752	75860	Obstipatie
A04.00	Weakness/tiredness general	721562		Moeheid/zwakte
H81.00	Excessive ear wax	715031	374375	Overmatig cerumen
S03.00	Warts	671915	140641	Wratten
L15.00	Knee symptom/complaint	623705		Knie symptomen/klachten

Column mapping

Source code column

Code

Source name column

English term

Source frequency column

Count

Auto concept ID column

UMLS lookup

Additional info column

Dutch term

Filters

☒ Filter by user selected concepts
☐ Filter by concept class: 2-dig nonbill code

☒ Filter standard concepts
☐ Filter by vocabulary: ABMS

☒ Include source terms
☒ Filter by domain: Condition

Cancel

Import

Import codes from ICPC2SNOMED.csv

Code	English term	Count	UMLS lookup	Dutch term
A99.00	General disease	5774012		Andere gegeneraliseerde/niet gespecific...
K86.00	Hypertension uncomplicated	3987206		Essentiële hypertensie zonder orgaan...
R44.00	Preventive Immunisations/Medications	3702922		Immunisatie/preventieve medicatie
T90.02	Diabetis mellitus type 2	2275799		Diabetes mellitus type 2
R05.00	Cough	1268829	4158493	Hoesten
R74.00	Upper respiratory infection acute	1061504		Acute infectie bovenste luchtwegen
A29.00	General symptom/complaint other	1035167		Andere algemene symptomen/klachten
L03.00	Low back symptom/complaint	998249		Lage rugpijn zonder uitstraling [ex. L86]
U71.00	Cystitis/urinary infection other	970719		Cystitis/urinewegsinfecties
A60.00	Results Tests/Procedures	903897		Uitslag onderzoek/verrichting
R97.00	Allergic rhinitis	892467	257007	Hooikoorts/allergische rhinitis
A97.00	No disease	868585		Geen ziekte
R96.00	Asthma	859684		Astma
P76.00	Depressive disorder	856960	440383	Depressie
P06.00	Sleep disturbance	777549		Slaapeloosheid/andere slaapstoornis
R95.00	Chronic obstructive pulmonary disorder	764354	4188974	Emfyseem/COPD
S88.00	Dermatitis contact/allergic	736689		Contact eczeem/ander eczeem
R78.00	Acute bronchitis/bronchiolitis	728202	4110022	Acute bronchitis/bronchiolitis
D12.00	Constipation	723752	75860	Obstipatie
A04.00	Weakness/tiredness general	721562		Moeheid/zwakte
H81.00	Excessive ear wax	715031	374375	Overmatig cerumen
S03.00	Warts	671915	140641	Wratten
L45.00	Knee symptom/complaint	623705		Knie symptomen/klachten

Column mapping

Source code column: Code

Source name column: English term

Source frequency column: Count

Auto concept ID column: UMLS lookup

Additional info column: Dutch term

Filters

nonbill code

Filter

Inclu

tion

Need to tell Usagi which column is what

Cancel Import

Import codes from ICPC2SNOMED.csv

Code	English term	Count	UMLS lookup	Dutch term
A99.00	General disease	5774012		Andere gegeneraliseerde/niet gespecif...
K86.00	Hypertension uncomplicated	3987206		Essentiële hypertensie zonder orgaan...
R44.00	Preventive Immunisations/Medications	3702922		Immunisatie/preventieve medicatie
T90.02	Diabetes mellitus type 2	2275799		Diabetes mellitus type 2
R05.00	Cough	1268829	4158493	Hoesten
R74.00	Upper respiratory infection acute	1061504		Acute infectie bovenste luchtwegen
A29.00	General symptom/complaint other	1035167		Andere algemene symptomen/klachten
L03.00	Low back symptom/complaint	998249		Lage rugpijn zonder uitstraling [ex. L86]
U71.00	Cystitis/urinary infection other	970719		Cystitis/urinewegsinfecties
A60.00	Results Tests/Procedures	903897		Uitslag onderzoek/verrichting
R97.00	Allergic rhinitis	892467		Allergische rhinitis
A97.00	No disease	868585		
R96.00	Asthma	859684		
P76.00	Depressive disorder	856960		
P06.00	Sleep disturbance	777549		d/andere slaapstoornis
R95.00	Chronic obstructive pulmonary disorder	764354		OPD
S88.00	Dermatitis contact/allergic	736689		em/ander eczeem
R78.00	Acute bronchitis/bronchiolitis	728202		itis/bronchiolitis
D12.00	Constipation	723752		
A04.00	Weakness/tiredness general	721562		Moeheid/zwakte
H81.00	Excessive ear wax	715031	374375	Overmatig cerumen
S03.00	Warts	671915	14064	Wratten
L15.00	Knee symptom/complaint	623705		Knie symptomen/klachten

Can specify target domain and filter rules

Column mapping

Source code column	Code
Source name column	English term
Source frequency column	Count
Auto concept ID column	UMLS lookup
Additional info column	Dutch term

Filters

- ☒ Filter by user selected concepts
- ☒ Filter standard concepts
- ☒ Include source terms
- ☐ Filter by concept class: 2-dig nonbill code
- ☐ Filter by vocabulary: ABMS
- ☒ Filter by domain: Condition

Cancel Import

Overview table

Usagi - ICPC25SNOMED_mapped.csv

Status	Source code	Source term	Frequency	Dutch term	Match score	Concept ID	Concept name	Domain	Concept class	Vocabulary	Concept code	Standard con.	Parents	Children	Comment
Unchecked	A99.00	General dise...	5774012	Andere gege...	0.78	435569	Disorder of j...	Condition	Clinical Findi...	SNOMED	37156001	S	1	39	
Unchecked	K86.00	Hypertension...	3987206	Essentiële...	0.72	42709887	Hypertensive...	Condition	Clinical Findi...	SNOMED	449759005	S	1	8	
Unchecked	R44.00	Preventive Im...	3702922	Immunisatie...	0.41	4102757	Response to...	Condition	Clinical Findi...	SNOMED	28529003	S	1	0	
Unchecked	T90.02	Diabetes mell...	2275799	Diabetes me...	0.91	201826	Type 2 diabet...	Condition	Clinical Findi...	SNOMED	44054006	S	1	14	
Auto mapped	R05.00	Cough	1268829	Hoesten	0.58	4158493	C/O - cough	Condition	Clinical Findi...	SNOMED	272039006	S	2	0	
Unchecked	R74.00	Upper respir...	1061504	Acute infectie...	1.00	260427	Common cold	Condition	Clinical Findi...	SNOMED	82272006	S	1	0	
Unchecked	A29.00	General sym...	1035167	Andere alge...	0.57	4037322	C/O: a gener...	Condition	Clinical Findi...	SNOMED	162409008	S	1	1	
Unchecked	L03.00	Low back sy...	998249	Lage rugpijn	0.62	4155085	C/O - a back...	Condition	Clinical Findi...	SNOMED	272009001	S	2	0	
Unchecked	U71.00	Cystitis/urina	970719	Cystitis/urine	0.54	81902	Urinary tract i...	Condition	Clinical Findi...	SNOMED	68566005	S	2	17	

Selected mapping

Source code	Source term	Frequency	Dutch term
A99.00	General disease	5774012	

Concept ID	Concept name	Domain	Concept class	Vocabulary
435569	Disorder of jaw	Condition	Clinical Finding	SNOMED

Search facility

Search Query

☒ Use source term as query

☐ Query:

Filters

☐ Filter by user selected concepts

☒ Filter standard concepts

☒ Include source terms

☐ Filter by concept class: 2-dig nonbill code

☐ Filter by vocabulary: ABMS

☐ Filter by domain: Condition

Results

Score	Term	Concept ID	Concept name	Domain	Concept class	Vocabulary	Concept code	Standard concept	Parents	Children
0.78	Oral disease	435569	Disorder of jaw	Condition	Clinical Finding	SNOMED	37156001	S	1	39
0.75	Venereal disease	440647	Sexually transmitted...	Condition	Clinical Finding	SNOMED	8098009	S	1	18
0.75	venereal disease	433967	Spirochetal infection	Condition	Clinical Finding	SNOMED	41116009	S	1	15
0.67	Viral disease	440029	Viral disease	Condition	Clinical Finding	SNOMED	34014006	S	1	36
0.64	Nil disease	4260398	Minimal change dis...	Condition	Clinical Finding	SNOMED	44785005	S	1	3
0.64	nil disease	199071	Nephrotic syndrom...	Condition	Clinical Finding	SNOMED	266549004	S	1	0
0.62	Viral diseases	140020	Viral exanthem	Condition	Clinical Finding	SNOMED	49882001	S	2	7
0.61	Genetic disease	443916	Hereditary disease	Condition	Clinical Finding	SNOMED	32895009	S	1	34
0.59	Degenerative disc d...	80816	Degeneration of int...	Condition	Clinical Finding	SNOMED	77547908	S	3	5

Comment:

Approved / total: 0 / 1714 0.0% of total frequency

Replace concept Add concept Approve

DBT

Transformación a OMOP-CDM

Sources

```
version: 2
```

```
sources:
```

```
- name: raw
```

```
  schema: raw
```

```
  tables:
```

```
    - name: 0_990_Complete-filtered_COMP_LABS_OLIGOCLONAL
```

```
    - name: 1_990_Complete-filtered_CONFIRMED BY
```

```
    - name: 2_990_Complete-filtered_Classification
```

```
    - name: 3_990_Complete-filtered_Clinical_Attacks
```

```
    - name: 990_Complete-filtered_H_W
```

```
    - name: 7_990_Complete-filtered_MRI_ACTIVITY
```

```
    - name: 990_Complete-filtered_OCB
```

```
    - name: 9_990_Complete-filtered_Order_med
```

```
    - name: 10_990_Complete-filtered_Patient
```

```
    - name: 11_990_Complete-filtered_Physical Exam Metrics
```

```
    - name: 12_990_Complete-filtered_RxNorm
```

```
    - name: 13_990_Complete-filtered_SDE_DMT
```

```
    - name: 14_990_Complete-filtered_Smoking
```

```
    - name: 15_990_Complete-filtered_Symptoms
```

```
    - name: 16_990_Complete-filtered_row19-Diagnosis Date
```

```
# - name: 6_990_Complete-filtered_Interval History
```

```
# - name: 4_990_Complete-filtered_ENCOUNTERS
```

Models

```
{{ config(
    materialized='table'
) }}

select
    "PERSON_ID",
    "PAT_ID" pat_id,
    "LINE",
    "SFGROUP",
    "MEDICATIONS" medication,
    "MED START DATE"::DATE med_start_date,
    "MED END DATE"::DATE med_start_date,
    "MED DISCONTINUATION REASON" med_discontinuation_reason
from {{ source('raw', '13_990_Complete-filtered_SDE_DMT') }}
```

Tables

Source

10_990_complete-ph
iltered_patient.csv

CDM_RiaH_v2.5

person

Fields

Source

*person_id

*pat_id

*birth_date

*sex

*country_lives_in

ethnic_group

race

pat_status

death_date

emp_status

CDM_RiaH_v2.5

*person_id

birthdate

*sex

country

ethnicity

race

death_status

death_date

employment_status

employment_type

Models

```
with patient as (  
    select *  
    from {{ ref('raw_patient') }}  
)
```

Models

```
SELECT  
    patient."PAT_ID" AS person_id,  
    patient."BIRTH_DATE" AS birthdate,  
    {{ sex_mapping('patient."SEX"') }} AS sex,  
    {{ country_mapping('patient."COUNTRY_LIVES_IN"') }} AS country,  
    {{ ethnicity_mapping('patient."ETHNIC_GROUP"') }} AS ethnicity,  
    {{ race_mapping('patient."RACE"') }} AS race,  
    {{ death_status_mapping('patient."PAT_STATUS"') }} AS death_status,  
    patient."DEATH_DATE" AS death_date,  
    {{ employment_status_mapping('patient."EMP_STATUS"') }} AS employment_status,  
    {{ employment_type_mapping('patient."EMP_STATUS"') }} AS employment_type,  
    NULL::TEXT AS inactivity_reason,  
    NULL::TEXT AS education_level,  
    {{ smoking_status_mapping('patient."SMOKING_STATUS"') }} AS smoking_status,  
    patient."PAT_ID" AS person_id_raw_value,  
    patient."BIRTH_DATE" AS birthdate_raw_value,  
    patient."SEX" AS sex_raw_value,  
    patient."COUNTRY_LIVES_IN" AS country_raw_value,  
    patient."ETHNIC_GROUP" AS ethnicity_raw_value,  
    patient."RACE" AS race_raw_value,  
    patient."PAT_STATUS" AS death_status_raw_value,  
    patient."DEATH_DATE" AS death_date_raw_value,  
    patient."EMP_STATUS" AS employment_status_raw_value,  
    patient."EMP_STATUS" AS employment_type_raw_value,  
    NULL::TEXT AS inactivity_reason_raw_value,  
    NULL::TEXT AS education_level_raw_value,  
    smoking."SMOKING_STATUS" AS smoking_status_raw_value  
FROM patient
```

Macros

```
--- Macro to transform 'H' and 'M' sex values into their concept_id
{% macro sex_mapping(sex) %}
(CASE WHEN {{ sex }} = 'MALE' THEN 'Male' -- man
      WHEN {{ sex }} = 'FEMALE' THEN 'Female' -- woman
      ELSE NULL::TEXT -- unknown
      END)
{% endmacro %}
```

```

with rxnorm as (
  select medication, rxnorm_code
  from {{ ref('sde_dmt_rxnorm') }}
)

```

Seeds

stg_symptom.sql

! sources.yml

> profile

seeds

.gitkeep

sde_dmt_rxnorm.csv

> snapshots

> target

> tests

.gitignore

! dbt_project.yml

≡ penn.db

ⓘ README.md

> RocheToCDM

> rts-client

> site-etl-pipeline

> streamlit-drag-and-drop

> tmp

> whiterabbit-merge

> whiterabbit-to-datadict

allow.bat

cocoon-main.zip

≡ GITHUB_PAT.txt

openapi-generator-cli-6.0.0.jar

	column 1	column 2
1	medication	rxnorm_code
2	Ocrevus	1876366
3	Copaxone (40mg)	214582
4	Tecfidera (240mg)	1373490
5	Kesimpta	712566
6	Avonex	75917
7	Copaxone (20mg)	214582
8	Gilenya	1012892
9	Tysabri	354770
10	Aubagio	1310520
11	Betaferon	72257
12	Rebif (44)	75917
13	Mavenclad	44157
14	Rituxan (every 6 months)	121191
15	Zeposia	2288236
16	Vumerity	2261783
17	Mayzent	2121085
18	Tecfidera (120mg)	1373479
19	Plegridy	1546168

Profile

```
penn:
  target: dev
  outputs:
    penn:
      type: postgres
      threads: 16
      host: "{{ env_var( 'HOST' ) }}"
      port: "{{ env_var( 'PORT' ) }}"
      user: "{{ env_var( 'USER' ) }}"
      pass: "{{ env_var( 'PASS' ) }}"
      dbname: "{{ env_var( 'DBNAME' ) }}"
      schema: "{{ env_var( 'SCHEMA' ) }}"

  dev:
    type: sqlite
    threads: 1
    database: 'penn.db'
    schema: cdm
    schemas_and_paths:
      main: 'penn.db'
    schema_directory: 'models'
```



Command	Description
dbt seed	Loads data from CSV files into your data warehouse. These CSV files are typically located in the data directory and are helpful for small, static datasets.
dbt compile	Compiles the dbt models into SQL files without running them against the database. Useful for checking that models are correctly set up.
dbt run	Executes the compiled models in the data warehouse, transforming raw data according to the specified models.
dbt docs	Generates documentation for the dbt project, including details about models, lineage, and dependencies.

Extras

Transformación a OMOP-CDM

PROFILING

ydata-profiling

patients Profiling Report

Overview Variables Interactions Correlations Missing values Sample

Overview

Brought to you by [YData](#)

Overview

Alerts 21

Reproduction

Dataset statistics

Number of variables	28
Number of observations	111
Missing cells	454
Missing cells (%)	14.6%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	24.4 KiB
Average record size in memory	225.2 B

Variable types

Text	21
Numeric	7

MAPPING using LLMs

Mapping of oncology regimen data through an LLM-enhanced pipeline

Tatsiana Skuhareuskaya¹, Mikita Salavei¹, Qi Yang², Maria K
1. Odysseus, an EPAM Company;
2. Analyst, Data Strategy, Access, and Enablement (DSAE), IQVIA

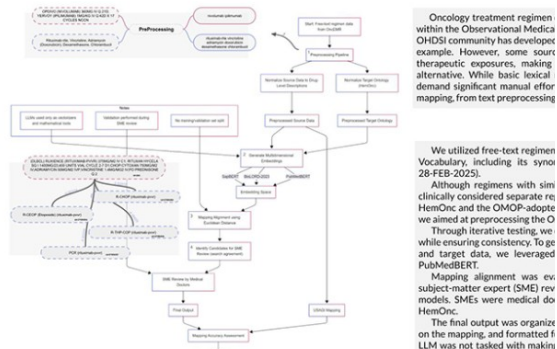


Fig.1 - Schematic approach for mapping and validation (with examples)

Results

After comparing the results yielded by Cosine similarity and Euclidean distance metrics, the decision was made to use the latter, since distances measured with Cosine similarity were smaller and thus were less reliable when choosing the mapping candidates. Our approach mapped 19,128 distinct codes to 960 unique HemOnc regimens, while effectively filtering out supportive care and irrelevant ("junk") codes, resulting in 2,735 codes with no corresponding regimen (zero matches). Examples of vectorizer input as well as the embedding grouping model used are presented in Figure 1.

Mapping accuracy achieved 79.6% reliability for cases where the candidate target was suggested by all three embedding models. For only two models, the LLM success rate was 20.4%.

Review of the results has shown the less prevalent cancer type is, the more reliable the mapping got. We presume this is the result of more stable regimens being present in the medical literature for a longer period of time, thus resulting in a bigger corpus of text the models were trained on.

We also compared the mappings suggested by the vectorization approach with those created by USAGI. For USAGI mappings, approximately 1 in 2 mappings had to be changed by the reviewer compared to 1 in 3 for the LLM-enhanced approach. Additionally, 362 regimens were mapped to a single corresponding nosology, highlighting their specificity.

We evaluated the mapping of source codes to HemOnc standard concepts across a range of rare and common tumors. Fig. 2 shows the number of source codes aggregated per target concept and the proportion of codes correctly suggested by the LLM, all with accuracy above 90%. Conditions with high LLM mapping accuracy may not require manual curation due to the high confidence in the automated suggestions (above-mentioned rare cancers).

We also examined the most frequent HemOnc cancer types the resulting mappings had a connection to. The top-20 conditions are represented in Fig. 3.

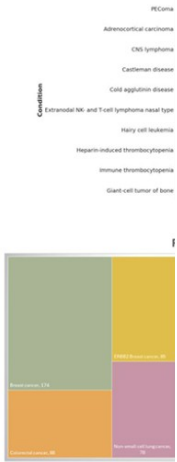


Fig. 3 - Top 20 cancer types

Conclusion

Semantic mapping of free-text oncology regimen data may be the only source of Episode related facts for many real world data sources help with reverse-engineering of Episode-event data. Our future work is focused on benchmarking algorithm-based tools used to derive community tools, i.e. Artemis.

Multimodal semantic search approach facilitates the efficient mapping of custom concepts in the intricate domain of treatment language models and exploration of general-use LLMs as compared to medically-trained ones, as well as introduction of new accurate improve precision.

EPAM IQVIA

Language-agnostic and expandable automated

concept mapping using semantic LLM agents

AI-Driven Precision: Semantic Search and Smart LLM Re-ranking
Mapping Croatian Medical Concepts to OMOP-CDM

Background: Mapping local medical terminologies to OMOP-CDM standard concepts is a time-consuming process, particularly for non-English data sources. Traditional manual mapping often lacks semantic accuracy, as direct translations frequently fail in medical contexts.

Methods

We developed a method inspired by retrieval-augmented generation (RAG) to leverage embeddings and intelligent reranking to preserve meaning and language nuances. The process involves a database to retrieve semantically similar concepts, which are then refined by an LLM-based reranking model. Additionally, the method generates confidence scores to enable acceptance of concepts where the model shows higher certainty.

- 1. Embeddings** Create dense vector representations of source codes and OMOP standard concepts
- 2. Vector Search** Retrieve top-k concepts based on vector similarity and metadata filtering
- 3. Reranking** Perform LLM-based reranking of retrieved concepts using domain-specific prompts and structured context
- 4. Validation** Enable interactive validation by medical experts through a custom-built web application

Results

92% precision
mapping drugs with unstructured descriptions

95% accuracy
on a sample of unique health interventions

Vector search + LLM > vector search only



Karlo Pintarić, MD, et al.
karlo.pintaric@hzjz.hr

Efficient automated mapping of internal source codes to OMOP CDM concepts

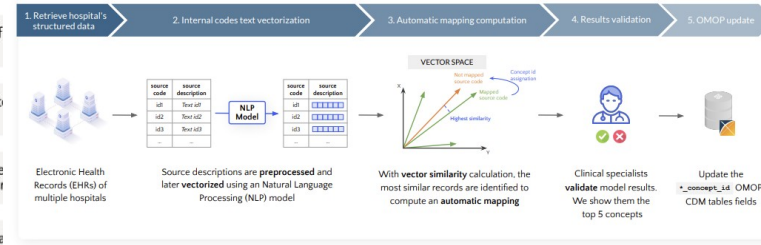
IO MED Medical Solutions

Automated Concept Mapping System for OMOP using Vector Representations and Cross-hospital Mapping

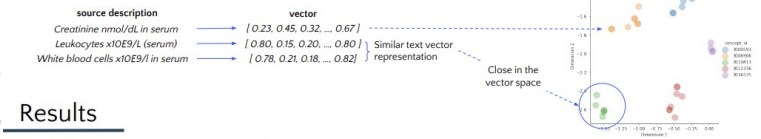
The searching based manual assignment of standard concepts in Observational Medical Outcomes Partnership (OMOP) to local source codes is a time-consuming and error-prone process. Although the OHDSI-developed tool USAGI was designed for this procedure, the non-English language data and OMOP integration limitations hinder efficient mapping.

Methods

We propose a method that leverages existing mappings from other hospitals, enabling the efficient scaling of a single mapping across all others. The mapping process is automated by computing vector representations of source code texts, which capture the relevant syntactic and semantic features, ensuring that similar records are grouped closely in the vector space. Consequently, assigning concept IDs becomes a matter of performing similarity searches within the vector space.



Example



Results

For model evaluation, a comprehensive dataset has been constructed:
- Data sourced from 11 distinct hospitals
- A total of 79,928 unique source codes

Accuracy 72%

Number of correct mappings executed by the model

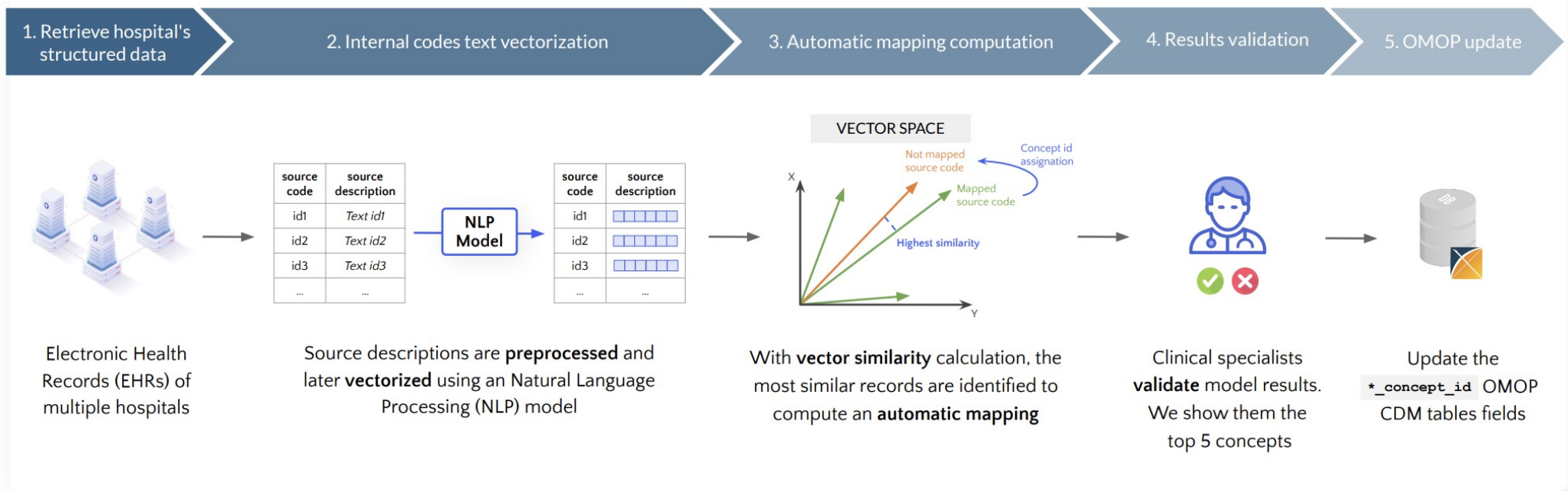
Top-5 Accuracy 88%

In our use case, shows how likely it is for the correct mapping to be amongst the ones provided to the clinical specialist

78.07% reduction in mapping validation time achieved

Figure. Text vector representations mapped to 5 concept IDs.



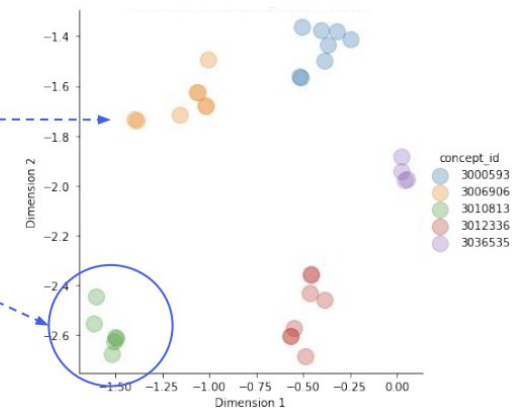


Example

source description	vector
Creatinine nmol/dL in serum	[0.23, 0.45, 0.32, ..., 0.67]
Leukocytes x10E9/L (serum)	[0.80, 0.15, 0.20, ..., 0.80]
White blood cells x10E9/l in serum	[0.78, 0.21, 0.18, ..., 0.82]

} Similar text vector representation

Close in the vector space





OMOP Concept Mapper

Contact

Keyword

type 2 diabetes

SEARCH

Save search

Found 50 concepts

< Prev

1

2

3

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ID	CODE	NAME	CLASS	CONCEPT	VALIDITY	DOMAIN	VOCAB	LINK	FAV
201826	44054006	Type 2 diabetes mellitus	Disorder	Standard	Valid	Condition	SNOMED		
45884697	LA15305-8	Type 2 diabetes (Adult onset)	Answer	Standard	Valid	Meas Value	LOINC		
4304377	81531005	Type 2 diabetes mellitus in obese	Disorder	Standard	Valid	Condition	SNOMED		
1076828	2222141000000103	Type 2 Diabetes in the Young review	Procedure	Standard	Valid	Observation	SNOMED		
4099651	190389009	Type 2 diabetes mellitus with ulcer	Disorder	Standard	Valid	Condition	SNOMED		
4230254	359642000	Type 2 diabetes mellitus in nonobese	Disorder	Standard	Valid	Condition	SNOMED		
1074568	2222151000000100	Type 2 Diabetes in the Young review offered	Context-dependent	Standard	Valid	Observation	SNOMED		
45757508	164971000119101	Type 2 diabetes mellitus controlled by diet	Disorder	Standard	Valid	Condition	SNOMED		

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AI Assistant

e.g., Map 'Type 2 Diabetes Mellitus' for 'condition_concept_id' in the 'condition_occurrence' table

Type 2 diabetes mellitus

Field

Concept ID

Code

Vocabulary

Domain

Class

Type

Validity

Athena

Type in the medical keyw

AI can make m

<https://www.omapper.org/>

cURL

Python

JavaScript

cURL (with details)

Python (with c

```
import requests

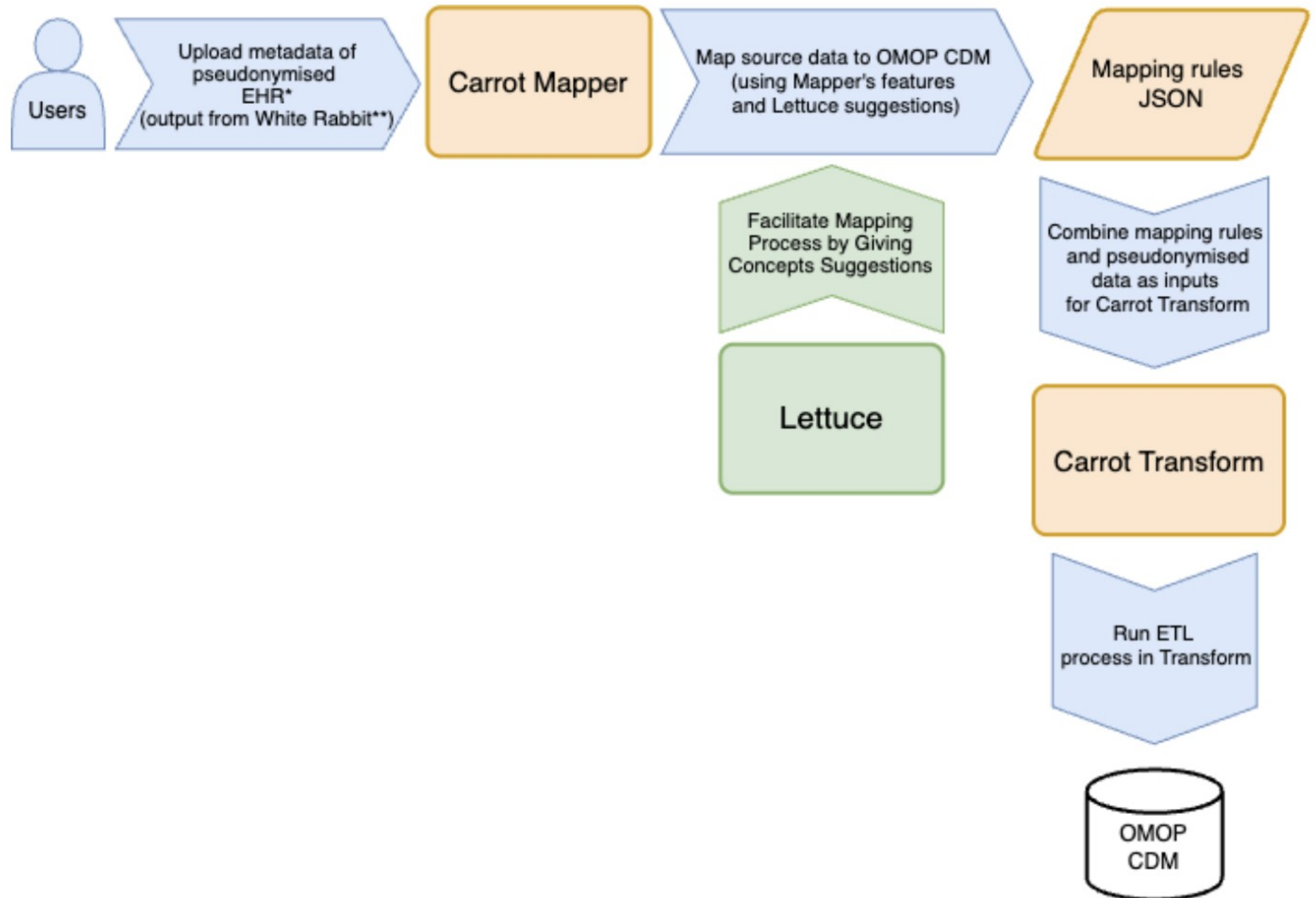
response = requests.get(
    "https://api.omophub.com/v1/concepts/320128",
    headers={"Authorization": "Bearer YOUR_API_KEY"}
)

concept = response.json()
```

```
{
  "success": true,
  "data": {
    "concept_id": 320128,
    "concept_name": "Essential hypertension",
    "concept_code": "59621000",
    "vocabulary_id": "SNOMED",
    "domain_id": "Condition",
    "concept_class_id": "Clinical Finding",
    "standard_concept": "S",
    "valid_start_date": "1970-01-01",
    "valid_end_date": "2099-12-31",
    "invalid_reason": null,
    "is_valid": true,
    "is_standard": true,
    "is_classification": false,
    "synonyms": [
```

<https://omophub.com/>





Preguntas?

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